

Chapter 157

Resolve FX Film Emulation

These plugins are designed to emulate motion picture film characteristics for aesthetic and creative effects.

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Camera Shake (Studio Version Only)

This effect replicates random camera motion for Horizontal and Vertical position, Rotation, and Zoom. A variety of parameters are provided to let you customize the quality of motion, from slow meandering drift to abrupt, jagged jerks and pops.

Main Controls

The primary controls governing how much camera shake is added.

- **Motion Scale and Speed Scale:** These sliders let you adjust the overall amplitude and speed of camera shake that's introduced to the image.
- **Motion Blur:** Lets you add simulated motion blur to the image being shaken, which can add a more realistic look to the motion. Motion blur is more visible on fast-moving motion, as DaVinci Resolve is simulating the effect of a motion picture camera's shutter speed on exposed subjects in motion.

Shake Levels

These controls let you adjust the intensity of camera shake.

- **Pan Amplitude:** Lets you set how much horizontal motion you want.
- **Tilt Amplitude:** This slider lets you choose how much vertical motion you want. These sliders are completely independent of one another.
- **Rotation Amplitude:** Lets you introduce rotational motion to the camera shake as well.
- **PTR Speed:** This slider lets you choose how fast the Pan, Tilt, and Rotation camera shake is.
- **Zoom Amplitude:** This slider lets you add zooming to the camera motion being created.
- **Zoom Speed:** This slider lets you choose how fast this random zooming should occur.
- **Zoom Type:** This pop-up menu lets you choose how to add zooming; you can choose Outward Only, Inward Only, or Outward and Inward. If you choose Inward Only, you don't have to worry about blanking appearing around the edges of the frame.

Shake Quality

These controls let you customize the type of shaking you want.

- **Motion Method:** This pop-up menu lets you choose from different ways of generating the motion this plugin introduces. You can choose from SINE, Rectified SINE, Rectified SINE (Invert), and Square Wave (good for popping or flickering motion).
- **Phase:** This slider lets you set the starting point of the camera shake you're creating, based on the Motion Method and Amplitude settings you've chosen.
- **Randomness Scale:** This slider lets you introduce irregularity to the Horizontal, Vertical, and Rotational motion of the camera shake. The greater this value, the more irregularity will be introduced.
- **Randomness Speed:** This slider lets you choose between smoothly erratic motion (at lower values) or more jagged motion (at higher values).
- **Pause Length:** This slider lets you adjust the frequency of intermittent pauses that break up the random motion added by this filter.
- **Pause Interval:** This slider lets you adjust the duration of intermittent pauses that break up the random motion added by this filter.
- **Pause Randomness:** This lets you add a degree of randomness to the intervals that happen.
- **Random Seed:** This slider lets you alter the value that sets what random values are being produced. Identical values result in identical randomness.

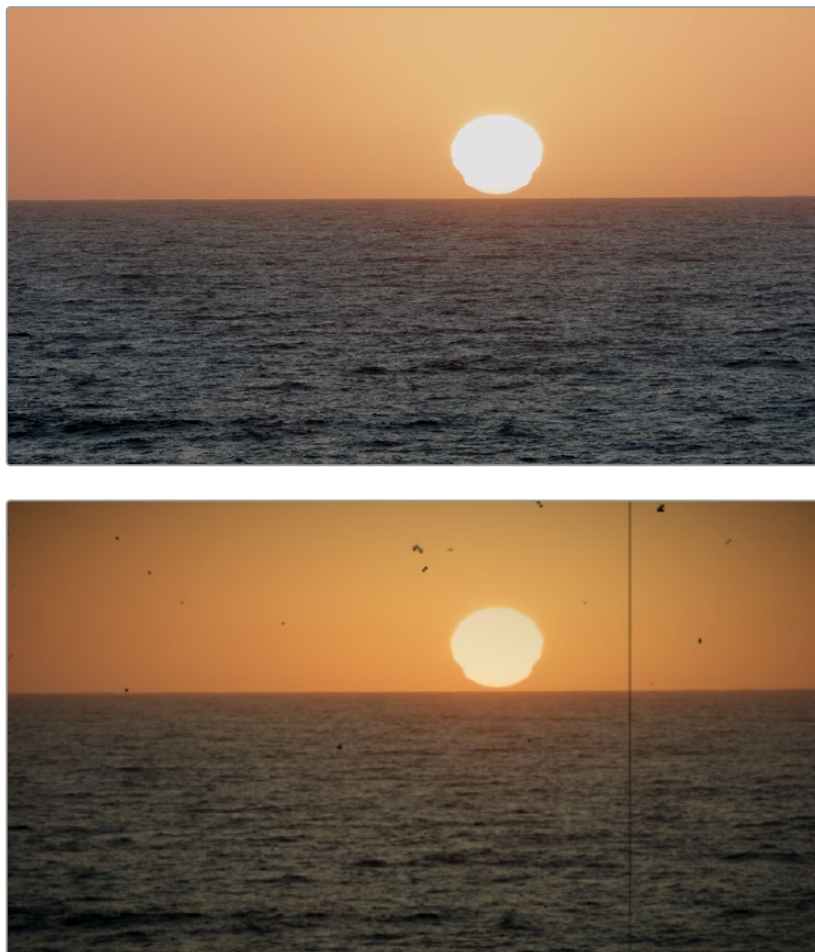
Blanking Handling

These controls let you determine what to do when camera shake causes blanking past the edge of the image.

- **Border Type:** This pop-up menu lets you choose how you want to fill the empty spaces at the edge that appear if the camera shake you create pushes the image out of the current frame size of your project. Four options let you choose to Replicate, Reflect, or Wrap-Around the image to fill the gap.
- **Zoom to Crop:** This slider lets you zoom in on the image to crop out unwanted blanking being introduced by the Camera Shake effect.

Film Damage

Found in the Resolve FX Texture category. After you've used the new Resolve FX Revival plugins to fix damage in archival footage, you can turn around and use the Film Damage plugin to make brand new digital clips look worn, dirty, and scratched instead. When used in conjunction with the Film Grain and Flicker Addition plugins, you can convincingly recreate the feel of poorly kept vintage archival footage.



(Top) Original image, (Bottom) Result with Film Damage

Blur and Shift Controls

The three parameters at the top let you alter the foundation of the image to begin creating the look of an older film.

- **Film Blur:** Adds just a bit of targeted defocusing to knock the digital sharpness out of the image.
- **Temp Shift:** Defaults to warming the image just a bit to simulate the warmer bulb of a film projector, although you can use it to cool or warm the image in varying amounts.
- **Tint Shift:** Defaults to yellowing the image to simulate damage to the film dyes, but you could move the slider in the other direction to add a bit of magenta, simulating a different kind of dye failure.

Add Vignetting

These parameters simulate lens vignetting darkening the edges of the image.

- **Focal Factor:** Adjusts how far the vignetting extends into the image.
- **Geometry Factor:** Affects how dark the vignetting is and how pronounced the edges are.
- **Tilt Amount:** Affects how balanced the vignetting at the top of the image is versus the bottom of the image.
- **Tilt Angle:** Affects how balanced the vignetting left of the image is versus the right, but only when Tilt Amount is set to something other than 0.

Add Dirt

These parameters let you simulate dirt particles (not dust) that have adhered to the film. These are larger specks, although there's several ways you can customize these.

- **Dirt Color:** This control lets you choose what color you want the dirt particles to be (black simulates dirt on a print, while white simulates dirt on a negative).
- **Changing Dirt:** This checkbox lets you alternate between simulating temporally unstable dirt on the film (checkbox on) and dirt on the lens that doesn't move (checkbox off).
- **Dirt Density:** Lets you choose more or fewer dirt particles appearing over time.
- **Dirt Size:** Lets you choose the average size of dirt particles to appear.
- **Dirt Blur:** Lets you defocus the dirt, so it's not so sharp.
- **Dirt Seed:** Changes the random distribution of dirt when you change this value, but for any given value, the results for any given set of control adjustments remain consistent.

Add Scratch

Five sets of Add Scratch parameters let you add up to five scratches to the image, simulating something scratching the emulsion while the film played.

- **Scratch Color:** Lets you choose the color you want the scratch to be (scratches can be a variety of colors depending on the depth of the scratch, type of film, and method of printing).
- **Scratch Position:** Lets you adjust the scratch's horizontal position on the image.
- **Scratch Width and Scratch Strength:** Let you adjust the scratch's severity.
- **Scratch Blur:** Lets you defocus the scratch.
- **Moving Scratch:** This checkbox lets you choose whether the scratch is jittering around or not.
- **Moving Amplitude:** Determines how far the scratch moves.
- **Moving Speed:** Determines how fast the scratch moves.
- **Moving Randomness:** Determines how the scratch meanders about.
- **Flickering Speed:** Determines how much the scratch flickers lighter and darker in severity.

Film Grain (Studio Version Only)

Composites a procedurally generated layer of simulated “film grain” over the image. Individual simulated grains with falloff are generated to match the resolution of the project.

Main Controls

These controls let you quickly choose and apply a film grain effect.

- **Film Grain Presets:** This pop-up lets you quickly choose from among different 8mm, 16mm, and 35mm film grain effects.
- **Composite Type:** Lets you choose a composite mode with which to combine the image with the grain simulated by this plugin.
- **Opacity:** Lets you fade the Film Grain effect against the original image.
- **Grain Only:** This checkbox lets you see just the grain layer that you’re creating, so you can evaluate its look prior to it being composited into the image.
- **Freeze:** Stops the motion of the grain animation, allowing you to use the effect for a still image or freeze frame.

Grain Params

The Grain Params group lets you adjust the quality of the grain.

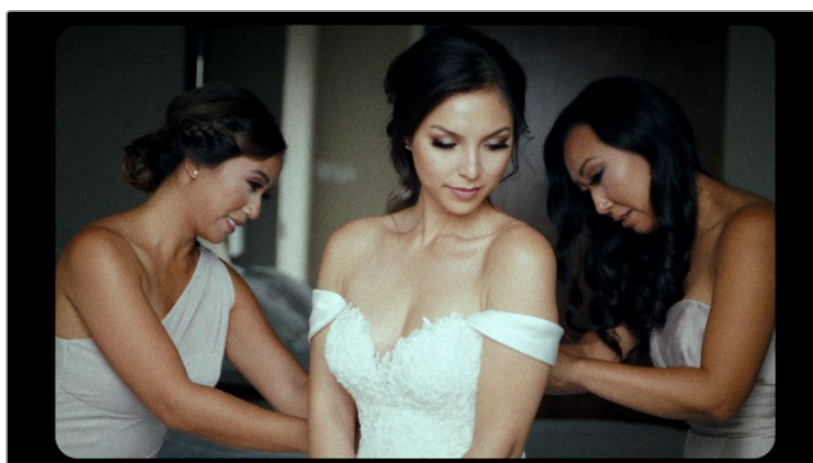
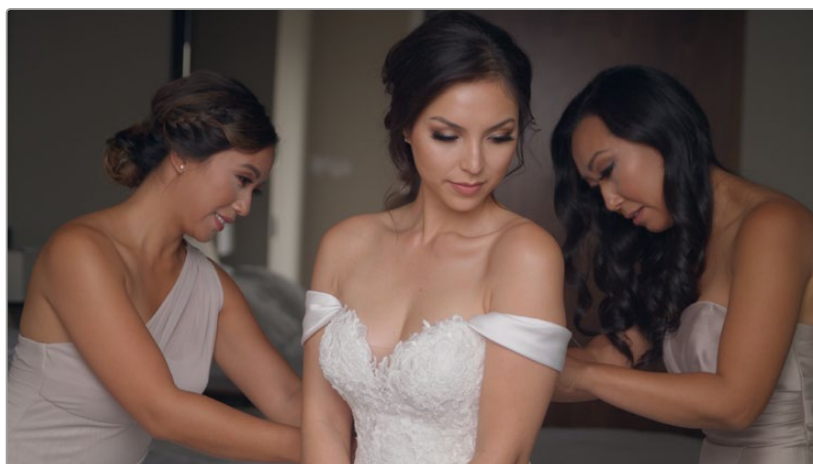
- **Texture:** Larger values create more separation between the grains generated by this plugin, while smaller values bunch them up closer together to create denser grain patterns.
- **Grain Size:** Larger values zoom into the individual simulated grains generated by this plugin, while smaller values zoom out.
- **Grain Aspect Ratio:** Lets you adjust the aspect ratio of the grains.
- **Grain Strength:** Adjusts the contrast of the simulated grain layer. Higher values produce more contrast, lower values less.
- **Offset:** Lightens or darkens the entire simulated grain layer. The practical effect is that lower values emphasize lighter grains, while higher values emphasize darker grains.
- **Symmetry:** An asymmetrical contrast adjustment. Lower values darken the lighter grains, while higher values brighten the darker grains.
- **Softness:** Lets you blur the grain layer to create even softer transitions from one grain to another.
- **Saturation:** Lets you make the grain more or less colorful. At a value of 0, grain is monochrome.

Advanced Controls

In the Advanced Controls, the Red, Green, and Blue sliders control the color of the grain. The Shadow, Midtone, and Highlight Gain controls let you adjust the amount of grain that’s composited into each of these tonal regions of the image. When the Optimized Mode checkbox is turned on (the default), performance should improve substantially. Turning on the Animate on Every Refresh checkbox will animate the grain as you adjust the sliders, allowing you a better idea of how the grain pattern will look in motion.

Film Look Creator (Studio Version Only)

The Film Look Creator gives you the ability to easily alter both the color and physical properties of an image to mimic motion picture film stock in a single Resolve effect. This effect is not focused on emulating a specific manufacturer's film stock, rather it creates a base "Blackmagic" film look that can be manipulated in a variety of creative ways. The Film Look Creator alters contrast, saturation, hue, and exposure of an image. It also includes the tools you want to have for film emulation including halation emulation, film gate, flicker, weave, vignetting, bloom, and more.



The original video (Top), and the Film Look Creator Bleach Bypass Super 16mm version (Bottom)

Many of the components of the Film Look Creator have specific counterparts in the Resolve FX library. However, the Film Look Creator combines these film look tools into a single Resolve effect with a greatly simplified tool set. If you ever find you need more control over an aspect of the look, you can always disable the effect's module and use a dedicated Resolve Effect in the node tree instead.

When this effect is applied, an underlying Blackmagic film look is automatically set first. This film look is driven by over 60 parameters that you get with the plugin active (and all effects off). These parameters are fine tuned and "baked in" to the Blackmagic film stock. But while these can't be adjusted, there are additional controls on the top area of the plugin that you can use to refine and create unique looks.

The Film Look Creator is designed for scene referred workflows and will work with any camera source and output display. When using either Resolve Color Management or ACES, this happens

automatically because the plugin uses the timeline color space for input and output by default. The plugin works for manually display referred timelines (such as DaVinci YRGB) as well, but then the user must manually choose the transform of the source footage and choose the display transform, which you can do within the plugin in the Color Space Overrides section.

NOTE: Film Look Creator works best with video that fills the entire frame. Effects like Halation and Film Gate may give unexpected results with video that is smaller than the frame size. If necessary, you can fill the frame using the Sizing controls, or in Project Settings > Image Scaling > Input/Output Scaling select “Scale Full Frame with Crop” rather than the default “Scale entire image to fit.”

Main Controls

These controls create the foundation of this effect.

- **Presets:** Automatically sets all controls for a desired preset look. “Clean Slate” does not apply the underlying Blackmagic Film Look and gives a clean image to work with. “Default No Effects” turns on the Blackmagic Film Look but disables all the additional effects below.
- **Color Blend:** Allows mixing the underlying color look and any changes made to other color settings and split tone controls.
- **Effects Blend:** The master control to adjust the strength of the spatial effects below. 0% is the effects completely off, and 100% is the full output of the Film Look Creator.
- **3D LUT Compatible:** Checking this box will remove spatial effect options which aren’t represented in a 3D LUT. This allows you to export your custom film look to other programs and camera hardware via a LUT.

Color Space Overrides

These controls allow you to change the input and output gammas and color spaces of the effect if you need to manually configure them for a display referred workflow. Please note that no tone mapping is applied if you change these. The default is “Use timeline,” which will match the color spaces you’ve selected in the Color Management section of the Project Settings.

- **Input Color Space:** Sets the input color space; choose one from the drop-down menu.
- **Input Gamma:** Sets the input gamma; choose one from the drop-down menu.
- **Output Color Space:** Sets the output color space; choose one from the drop-down menu.
- **Output Gamma:** Sets the output gamma; choose one from the drop-down menu.
- **Output White Point:** Sets the display white point. For creative purposes, you can choose one from the drop-down menu. D65 is the default white point.

Film Look

These controls allow you to adjust the overall parameters of the base film look.

- **Film Look Blend:** Adjusts the strength of the initial default base film look, before other parameters are adjusted.
- **Skin Bias:** Adjusts the skin tones of the film look. Negative values provide warm, darker, and more saturated skin tones, while positive values provide brighter, rosier, and less saturated skin tones.

Color Settings

These controls allow you to change the color information for the film look. These effects will be passed in a 3D LUT.

- **Exposure:** A photometric control that adjusts the exposure in stops (+/- 5).
- **Contrast:** Adjusts the overall S-curve transfer function to add or remove contrast from the shot.
- **Highlights:** Adjusts the roll-off at the top of the S-curve, allowing you to reign in or expand the brightest areas of the shot.
- **Fade:** Adjusts the roll-off at the bottom of the S-curve, allowing you to reign in or expand the darkest areas of the shot.
- **White Balance:** This control uses chromatic adaptation to mimic what the source footage might look like if it were shot under different lighting conditions. Adjusted in degrees Kelvin.
- **Tint:** Lets you adjust the green/magenta balance of the clip for balancing footage shot under limited spectrum lighting such as fluorescent, or sodium vapor streetlights.
- **Skin Bias:** Lets you adjust the skin tones of the image. From -1 (warmer, darker, more saturated) to +1 (rosy, brighter, and less saturated). This can also be useful if you may, for example, push tint in one direction, and then use Skin Bias to balance out the image.
- **Subtractive Sat:** Allows adjusting overall image saturation in a way that more closely resembles a subtractive method.
- **Richness:** Affects already more saturated areas, applying subtractive saturation to just those regions. Can be useful for adding depth and richness, or pulling back a bit on more saturated areas.
- **Bleach Bypass:** Simulates this well-known look by adding a desaturated, higher contrast version of the image onto itself.

Split Tone

These controls add color to shadows and highlights using an opposing hue angle. For example, adding orange to the highlights, and teal to the shadows to get a more modern cinematic look. These effects will be passed in a 3D LUT.

- **Enable Split Tone:** Turns this effect on or off.
- **Amount:** The intensity of the split tone. This value must be greater than 0 to see an effect.
- **Hue Angle:** This slider adjusts what complimentary hues are applied to the shadows and highlights.
- **Pivot:** This control sets where the hue transition occurs in luma. Set to its extreme of 0.000 or 1.0000, the pivot lets you apply a single tint to the shadows or highlight.

Vignette

These controls are simplified window for adding a vignette to the shot. For more detailed control you could disable this setting and use the Vignette effect instead.

- **Enable Vignette:** Turns the vignette on or off.
- **Amount:** Sets the intensity of the darkened areas of the vignette.
- **Size:** Sets how much the vignette reaches into the frame.

Halation

These controls are a simplified way of adding soft glows to the edges of the subjects in your frame, to mimic this characteristic of motion picture film. If you want more control, you can disable this setting and use the Halation effect instead.

- **Enable Halation:** Turns the Halation effect on or off.
- **Highlights Only:** Limits the Halation effect only to the edges of highlights.
- **Amount:** Lets you set the intensity of the halation.
- **Radius:** Lets you set the spread size of the glow from the edges.
- **Saturation:** Lets you adjust the color intensity of the halation.
- **Hue:** Lets you tint the glow a selected color.

Bloom

These controls lets you introduce a subtle glow around the edges of highlights.

- **Enable Bloom:** Turns the Bloom effect on or off.
- **Amount:** Lets you set the intensity of the bloom.
- **Radius:** Lets you set the spread size of the bloom from the edges.

Grain

These controls are a simplified set of tools to emulate the grain patterns found on motion picture film. For more detailed control you can disable this setting and use the Film Grain effect instead.

- **Enable Grain:** Turns the grain pattern on or off.
- **Preset:** Provides some starting points for grain, based on film stock standards.
- **Amount:** Determines the amount of grain in the image.
- **Size:** Determines the size of the grain in the image.
- **Softness:** Determines how crisp the individual grains are.
- **Saturation:** Adds color to the grain. 0 is monochromatic.
- **Image Defocus:** Affects the overall resolution of the image, defocusing more as you reduce the control towards 0. At 1.000 the resolution is untouched. This control does not affect the grain.

Flicker

These controls add a gentle pulse to the image exposure, mimicking the flicker of motion picture projection. For more detailed controls you can disable this setting and use the Flicker Addition effect instead.

- **Enable Flicker:** Turns the Flicker effect on or off.
- **Amount:** Sets how much flicker to add to the image. Low settings will be barely visible, while high settings can be useful for more extreme looks like the Nostalgic preset.
- **Rate:** The rate at which the flicker pulses.

Gate Weave

These controls mimic the slight horizontal and vertical shift of the image as the film is mechanically pulled through the camera's gate.

- **Enable Gate Weave:** Turns the Gate Weave on or off.
- **Amount:** Determines the amount of motion to the weave.
- **Rate:** The rate at which the weave moves.

Film Gate

These controls will add blanking to the edges of the image that mimic film projection or telecine alignment controls.

- **Enable Film Gate:** Turns the Film Gate on or off.
- **Preset:** Select an aspect ratio from different common film standards.
- **Ratio H/V:** You can set a custom ratio for the image here.
- **Enable Curvature:** Turning this on slightly rounds the corners of the image to mimic how the shape of an actual film frame is exposed.
- **Padding:** This will zoom the image in and out of the blanking, allowing you to see the entire exposed "frame" of film if desired. Similar to a telecine control.

NOTE: It is possible to get unexpected results using Film Gate with video that is not scaled to fill the entire frame. If this is the case, use the sizing controls to fill the frame with your image first.

Flicker Addition

Why remove flicker with the Resolve FX Flicker Removal plugin when you can add it instead? The Flicker Addition plugin adds rapidly animated exposure changes to make the image appear to flicker, creating animated effects that would be difficult to keyframe manually. When applied to an image in different ways, this plugin can be used to simulate torchlight, firelight, light fixtures with old ballasts or frayed wiring, or any temporally unstable light source. For example, you could key only the highlights of a night-time image and use Flicker Addition to affect those isolated highlights.

Two groups of controls let you control the quality of this flickering.

Main Controls

These controls let you choose how to apply the flicker and its overall speed and intensity.

- **Flicker Type:** This pop-up menu lets you apply the flicker as a Lift, Gamma, Gain, Vignette, or Alpha adjustment.
- **Range slider:** This lets you set how widely the flickering will vary.
- **Speed:** This lets you adjust how quickly the flickering is animated.
- **Smoothness:** This slider lets you adjust the temporal quality of the flickering, whether it changes abruptly from one value to another (at lower settings) or whether it makes more continuous transitions from one value to another (at higher settings).
- **Flicker R, G, B Channels:** Three checkboxes let you choose which color channels are affected by this flickering.

Flicker Quality

These controls let you adjust the details of how the flickering animates.

- **Randomness Scale:** This slider lets you introduce irregularity to the Horizontal, Vertical, and Rotational motion of the camera shake. The greater this value, the more irregularity will be introduced.
- **Pause Length:** This slider lets you adjust the frequency of intermittent pauses that break up the random motion added by this filter.
- **Pause Interval:** This slider lets you adjust the duration of intermittent pauses that break up the random motion added by this filter.
- **Pause Randomness:** This lets you add a degree of randomness to the intervals that happen.
- **Random Seed:** This slider lets you alter the value that sets what random values are being produced. Identical values result in identical randomness.

Halation (Studio Version Only)

An effect that mimics the subtle light scatters, reflections, and analog blooms of light reflecting back through the dye layers of motion picture film. This usually presents itself as a fine red/orange tinted glow around high contrast bright regions. Halation can also be caused by light reflections within processing and development equipment, or in the camera itself. These minor flaws can add a subtle organic and analog look to digital files.



The Halation effect can add an organic and analog look to your digital files. Original (L), Halation (R).

There are a variety of tools and controls to give you fine detail on exactly how the halation will appear.



The shot before the Halation effect



The Halation effect on the same image; note the halation in the water reflections and the woman in the white t-shirt.

Processing Color Space

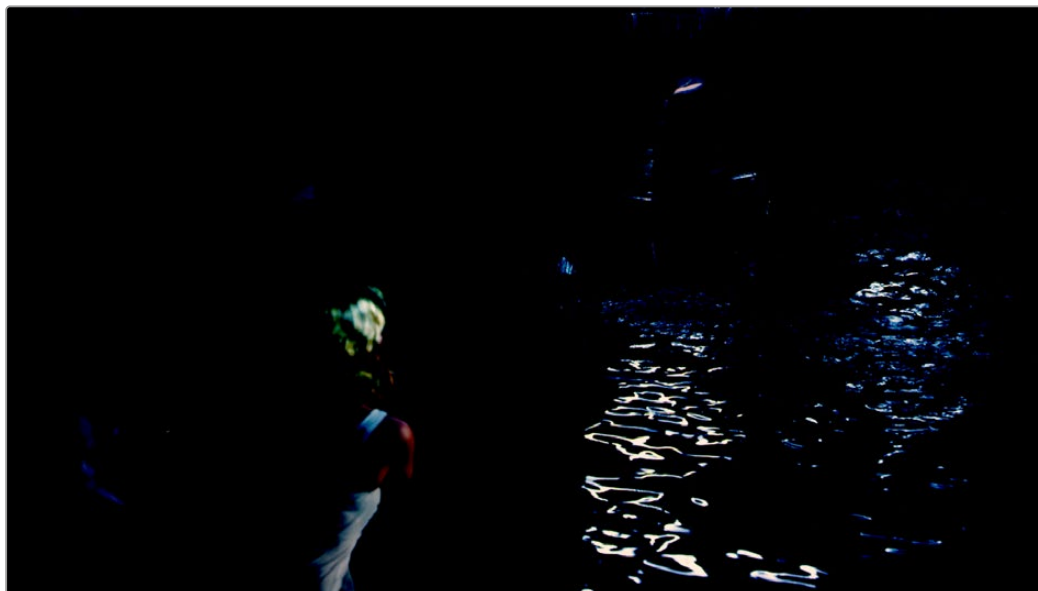
Lets you chose the color space for the Halation effect. The default is the same as the current timeline.

Isolation

The Isolation controls define which regions of your clips will form halation halos.

- **Threshold:** The level low clip level for the Halation effect.
- **Normalization:** The high clip level for the Halation effect.

- **Film Saturation Level:** The level above which all color is suppressed and saturated to white. Especially useful in HDR workflows, so the resulting Halation effects are visible at HDR brightness levels.
- **View Isolated Regions:** Shows only the regions isolated for generating the Halation effect.



The View Isolated Regions checkbox shows you the areas that will be affected by the halation.

Dye Layer Reflections

The Dye Layer Reflections controls give you the ability to fine tune the physical characteristics of the halation glow.

- **Strength:** Controls the brightness of the reflection.
- **Gamma:** Controls the how the glow spreads out.
- **Saturation:** Controls the intensity of the color of the glow.
- **Spread:** Defines the extent of the dye reflection.
- **Fine Tune Relative Spread:** Checking this box allows you to manually adjust the red, green, and blue reflection distances.

Secondary Glow

The Secondary Glow controls add additional halation characteristics.

- **Strength:** Controls the brightness of the secondary glow.
- **Gamma:** Controls how the glow spreads out.
- **Spread:** Defines the extent of the glow around bright detail.
- **Filter:** Limits the color of the glow in the selected color.

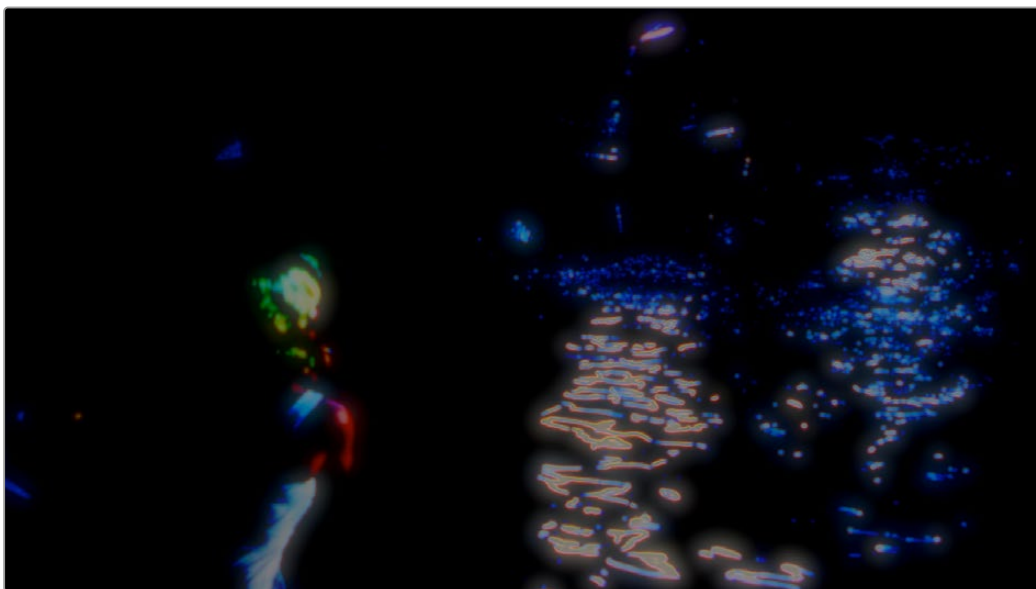
Basic Grain

These controls are a subset of the Film Grain effect for quick grain settings. If you prefer, you can add an instance of the full Film Grain effect after the Halation effect on the same node for greater detail.

- **Append Gain internally:** Turns on the simplified grain control.
- **Strength:** Adds grain to the halation layer.
- **Size:** Controls the size of the grains in the halation layer.
- **Softness:** Softens the grain texture.
- **Saturation:** Applies saturation on the grain.

Global Adjustments

- **View Glow Alone:** This checkbox shows the Halation effect alone before it's added to the frame. This is a useful setting to turn on when modifying any of the above tools.



The View Glow Alone checkbox shows you only the Halation effect in isolation and not blended with the underlying image.

- **Reduce Highlights:** Reduces the effect where it will brighten the scene without contributing to the halo. This essentially confines the halo to the edges of a region without blowing out the region itself.
- **Aspect Ratio:** Stretch or squeeze the glow and grain horizontally for anamorphic projects.
- **Detail Loss:** Removes sharp detail that the film stock should not have captured by defocusing the image under the effect.

Vignette

A plugin with two modes for creating vignette effects of different kinds.

- Basic mode gives you Size, Anamorphism, and Softness controls for quickly creating a traditional lens vignetting effect to darken the edges of the frame. A Color control lets you tint the vignette.
- Advanced mode adds Border Shape, Rotation, Center, Transparency, and Composite Type controls for customizing this vignetting effect even more specifically.